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SYSTEM AND METHOD FOR DIGITAL SCENT DELIVERY

Abstract

A system includes one or more vessels. Each of the vessels includes a scented medium between a first opening and a second opening. A first valve is disposed at the first opening of each of the one or more vessel. A second valve is disposed at the second opening of each of the one or more vessels. At least one of the first valve or the second valve of one of the one or more vessels is part of a first electromechanical actuation subsystem and another of the first valve or the second valve is a passive valve or is part of a second electromechanical actuation subsystem. A controller controls release of scented gas from at least one of the one or more vessels via the second opening to a corresponding outlet.

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Background/Summary

RELATED APPLICATIONS [0001] This application claims priority to and the benefit under 35 U.S.C. § 119(e) of U.S. provisional patent application No. 63/561,836, filed Mar. 6, 2024, entitled “SYSTEM AND METHOD FOR DIGITAL SCENT DELIVERY”, for which the entire contents are hereby incorporated by reference by their entirety herein.

SUMMARY

[0003] A system includes one or more vessels. Each of the vessels includes a scented medium between a first opening and a second opening. A first valve is disposed at the first opening of each of the one or more vessel. A second valve is disposed at the second opening of each of the one or more vessels. At least one of the first valve or the second valve of one of the one or more vessels is part of a first electromechanical actuation subsystem and another of the first valve or the second valve is a passive valve or is part of a second electromechanical actuation subsystem. A controller controls release of scented gas from at least one of the one or more vessels via the second opening to a corresponding outlet.

[0004] According to one aspect a system is provided. The system comprises a plurality of vessels, each of the plurality of vessels including a scented medium between a first opening and a second opening, a first valve disposed at the first opening of each of the plurality of vessels, a second valve disposed at the second opening of each of the plurality of vessels, wherein for each of the plurality of vessels, at least one of the first valve or the second valve is part of a first electromechanical actuation subsystem and another of the first valve or the second valve is a passive valve or is part of a second electromechanical actuation subsystem; one or more reservoirs holding a gas, one or more interconnections arranged to supply the gas from the one or more reservoirs to the first opening of the plurality of vessels, a controller configured to control release of scented gas from at least one of the plurality of vessels via the second opening of the at least one of the plurality of vessels to a corresponding outlet.

[0005] According to one embodiment, the scented gas results from the gas from one of the one or more reservoirs interacting with the scented medium in the at least one of the plurality of vessels. According to one embodiment, the scented medium is foam, wax, a paper substrate, gel, cotton ball, cotton pad, or porous polymer. According to one embodiment, the first electromechanical actuation subsystem and the second electromechanical actuation subsystem are a same type of electromechanical actuation subsystem. According to one embodiment, at least one of the first electromechanical actuation subsystem or the second electromechanical actuation subsystem includes a thermoelectric actuator. According to one embodiment, the thermoelectric actuator includes a shape memory alloy configured to move based on an application of a current and a stopper configured to cover at least one of the first opening or the second opening until movement of the shape memory alloy moves the stopper. According to one embodiment, the thermoelectric actuator includes Nitinol.

[0006] According to one embodiment, at least one of the first electromechanical actuation subsystem or the second electromechanical actuation subsystem operates based on an electrostatic or electrodynamic effect. According to one embodiment, at least one of the first electromechanical actuation subsystem or the second electromechanical actuation subsystem includes a piezoelectric actuator. According to one embodiment, the gas is pressurized air. According to one embodiment, the gas is liquified gas. According to one embodiment, the liquified gas is carbon dioxide. According to one embodiment, the system further comprising a fan or air pump coupled to the one or more reservoirs to urge flow of the gas from the one or more reservoirs. According to one embodiment, the controller is configured to communicate with an external device and control release of the scented gas based on communication from the external device. According to one embodiment, the controller is configured to communicate with the external device wirelessly or via a port. According to one embodiment, the system further comprising a cartridge configured to house the plurality of vessels. According to one embodiment, the system further comprising a cartridge interface, wherein the cartridge is configured to detach from the system and the cartridge interface is configured to couple to the cartridge when the cartridge is attached to the system.

[0007] According to one embodiment, the cartridge interface includes first openings configured to align, respectively, to the first openings of the plurality of vessels housed in the cartridge, and first gaskets configured to seal an interface between the first openings of the cartridge interface and the first openings of the plurality of vessels, respectively. According to one embodiment, the cartridge includes grooves defining each channel from the second opening of each of the plurality of vessels to each corresponding outlet and the cartridge interface includes channel gaskets configured to seal each of the channels. According to one embodiment, the cartridge includes a sheet covering a side of the cartridge that couples to the cartridge interface. According to one embodiment, each channel is between the sheet and one of the grooves. According to one embodiment, the sheet is a film or a laminate. According to one embodiment, the system further comprising a removable cover configured to cover the cartridge. According to one embodiment, the controller obtains information from the cartridge. According to one embodiment, the information identifies a scent stored in each of the plurality of vessels. According to one embodiment, the information is stored in an electrically erasable programmable read-only memory (EEPROM) of the cartridge. According to one embodiment, the information indicates allowed and disallowed combinations of scents. According to one embodiment, the information indicates usage of one or more of the plurality of vessels. According to one embodiment, the information is used to replace the cartridge. According to one embodiment, the information is used to refill or replace one or more of the plurality of vessels.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] Various aspects of at least one example are discussed below with reference to the accompanying figures, which are not intended to be drawn to scale. The figures are included to provide an illustration and a further understanding of the various aspects and examples, and are incorporated in and constitute a part of this specification, but are not intended as a definition of the limits of a particular example. The drawings, together with the remainder of the specification, serve to explain principles and operations of the described and claimed aspects and examples. In the figures, each identical or nearly identical component that is illustrated in various figures is represented by a like numeral. For purposes of clarity, not every component may be labeled in every figure. In the figures:

[0009] FIG. **1**A is a perspective view of aspects of a digital scent delivery device according to some embodiments;
[0010] FIG. **1**B is a perspective view showing different aspects of the digital scent delivery device shown in FIG. **1**A;
[0011] FIG. **2** is a perspective view showing a slidable cover of an exemplary digital scent delivery device according to some embodiments;
[0012] FIG. **3**A is a perspective view of an exemplary digital scent delivery device according to some embodiments with an outer cover removed;
[0013] FIG. **3**B shows the exemplary digital scent delivery device of FIG. **3**A with a cartridge of vessels detached;
[0014] FIG. **4** is a perspective view of an exemplary digital scent delivery device according to some embodiments with the housing removed;
[0015] FIG. **5** shows the exemplary digital scent delivery device of FIG. **4** with a controller cover removed;
[0016] FIG. **6** is a cross-sectional view of a first side of an exemplary digital scent delivery device according to some embodiments;
[0017] FIG. **7** is a cross-sectional view of a second side of the exemplary scent delivery device shown in FIG. **6**;
[0018] FIG. **8** is a block diagram showing an exemplary interface between a reservoir and a vessel **215** according to some embodiments;
[0019] FIG. **9** is a cross-sectional view of an exemplary digital scent delivery device according to some embodiments;
[0020] FIG. **10** is an expanded view of aspects of the cross-sectional view shown in FIG. **9**;
[0021] FIG. **11** is an expanded view of aspects of the cross-sectional view shown in FIG. **9**;
[0022] FIG. **12**A is a block diagram illustrating a channel for scent delivery according to one embodiment;
[0023] FIG. **12**B is a block diagram illustrating a channel for scent delivery according to one embodiment;
[0024] FIG. **12**C is a block diagram illustrating a channel for scent delivery according to one embodiment;
[0025] FIG. **13** is a block diagram of an exemplary vessel of according to some embodiments;
[0026] FIG. **14**A is a block diagram of a closed valve that is part of an actuation subsystem that uses thermoelectric effect according to exemplary embodiments;
[0027] FIG. **14**B is a block diagram of a closed valve that is part of an actuation subsystem that uses thermoelectric effect according to exemplary embodiments;
[0028] FIG. **15**A is a block diagram of the valve of FIG. **14**A in an open position;
[0029] FIG. **15**B is a block diagram of the valve of FIG. **14**B in an open position;
[0030] FIG. **16** is a block diagram of a closed valve that is part of an actuation subsystem that uses piezoelectric effect according to exemplary embodiments;
[0031] FIG. **17** is a block diagram of the valve of FIG. **16** in an open position;
[0032] FIG. **18** is a block diagram of a closed valve that is part of an actuation subsystem that uses electrostatic effect according to exemplary embodiments;
[0033] FIG. **19** is a block diagram of the valve of FIG. **18** in an open position;
[0034] FIG. **20** is a block diagram of a digital scent delivery device according to some embodiments; and
[0035] FIG. **21** is a block diagram of a digital scent delivery device according to some embodiments.

DETAILED DESCRIPTION

[0036] According to some implementations, a digital scent delivery device is provided that is capable of emitting scent to a user. The inventors appreciated the need for a commercially-available digital scent delivery device for controllably rendering scent in any number of applications. Scent may be added to digital experiences (e.g., games, art exhibits, shows) that may be audio, visual, or audio-visual. In some embodiments, the digital scent delivery device may be used directly with a device such as a mobile phone, gaming system or other device or system to render scent to a user. In other embodiments such as virtual reality (VR), augmented reality (AR), mixed reality (MR), or any of those environments (XR), scent may be released in coordination with other digital experiences. A variety of other applications may also benefit from the addition of a digitally actuated scent emitting device, which may be referred to as an olfactory device. For example, scent may be added to a two-dimensional media experience (e.g., television, movie, game) or an interactive environment (e.g., game, application). In the wellness space, a digital scent delivery device may be used for aromatherapy (e.g., to trigger scent to stimulate portions of the brain in a different way than audio or visual effects to address anxiety by promoting relaxation or to address other health issues) or in a training environment for enhanced realism.

[0037] According to some embodiments, a digital scent delivery device may be configured to render scent directly to a user. For example, the device may be placed close to a user (e.g., to a user's nose) and the amount of scent emitted may be limited such that a personal scent experience may be created without scent contamination among different users occupying the same space. Spatial features of the environment may be used to control which scent to emit and how much of the scent to emit. Scent delivery may be triggered by proximity, location, or activity, for example, in a three-dimensional environment (e.g., XR) or a two-dimensional representation. According to some embodiments, scent delivery may be based on spatial characteristics of odor impression in a game or an XR environment as described in U.S. Pat. No. 11,351,450. According to other embodiments, a digital scent delivery device may provide the same scent experience for a number of people. Passive and/or electromechanical actuation subsystems may be used to control emission of one or more scents while preventing leakage of other scents. In some applications, a scent delivery device may be attached, via brackets or another attachment mechanism, to another device (e.g., game component, VR component, any glasses or goggles, etc.) or may be worn (e.g., via a lanyard) or otherwise held near a person for scent delivery.

[0038] According to exemplary embodiments, an actuation subsystem may include a valve at a first (separable) portion of the digital scent delivery device and actuation elements at another (core) part of the device. A removable/replaceable cartridge may include vessels that hold the various scents. Each of the vessels may include a first opening for a medium (e.g., pressurized gas (e.g., air), liquified gas (e.g., carbon dioxide)) to enter the vessel and a second opening for scented medium to exit the vessel. This cartridge may be coupled to the digital scent delivery device, which includes a controller to control scent delivery. In some embodiments, a film or laminate on the cartridge may act as a valve on the first opening and/or second opening. The film or laminate may be attached to the cartridge via adhesive, ultrasonic or laser bonding, or heat sealing.

[0039] For explanatory purposes, a valve may be a component that contacts a medium and prevents or allows flow of the medium into or out of an opening, while actuation elements may be one or more components that affect the valve to control the flow of the medium into or out of the opening. In exemplary embodiments including the film or laminate acting as a valve on the first and/or second opening of the vessels of the cartridge, the actuation elements may be part of the digital scent delivery device to which the cartridge is coupled, as detailed. Because the valve (i.e., the component that contacts the medium) is part of the cartridge while the actuation elements, which do not contact the medium, are part of the core digital scent delivery device, the cartridge may be replaced with another cartridge with different scents without residual scent from the previously coupled cartridge contaminating the scents of the new cartridge.

[0040] FIG. **1**A is a perspective view of aspects of a digital scent delivery device **100** according to some embodiments. A first side **101** of the device **100** is visible in the view shown in FIG. **1**A. A separable cover **110** is shown covering a portion of the first side **101** of the device **100**. The cover

110 couples to a housing **115** of the device **100**, as further discussed with reference to FIG. **2**. The device **100** may communicate with any number and types of external devices to control emission of one or more scents. Non-limiting exemplary external devices that may communicate with and/or control scent delivery via the device **100** include an XR device, computer, cell phone, television, or gaming device. The external device may connect to the device **100** via a wire and connector to a port **105** (e.g., universal serial bus (USB) port) or may connect wirelessly (e.g., via Bluetooth or other protocol).

[0041] FIG. **1**B is a perspective view showing different aspects of the digital scent delivery device **100** shown in FIG. **1**A. A second side **102** of the device **100**, opposite the first side **101**, is visible in FIG. **1**B. The outlets **120** available to emit scent are shown. Generally, one or a subset of the outlets **120** may emit scent at a given time. According to some embodiments, multiple scents may be emitted in turn or a blended scent may be provided through simultaneous actuation of two or more scent emissions. An on/off button **130** is shown on the second side **102** of the device **100**. In alternate embodiments, any known power activation mechanism may be used. A light indicator **135** may emit light when the device **100** is turned on, for example. The light indicator **135** may emit different colors (e.g., green light when the device is on, red light when an error has occurred). Vents **140** for heat exhaust from aspects of the controller (e.g., processing circuitry and electronics, generally) are also shown on the second side **102** of the device **100**.

[0042] According to various non-limiting embodiments, the device **100** may be held by, worn by, or attached to a person to controllably provide a personal scent experience. For example, the device **100** may be pinned, adhered, or otherwise attached to the clothing of a user. The device **100** may also be worn (e.g., on a necklace, bracelet, watch, ring, head wrap, goggles), embedded in or connected to other devices (e.g., headphones, computer, keyboard) or may be handheld or placed on a surface. The interaction or portability of the device **100** is not intended to be limited by any of the examples noted herein. According to some embodiments, the device **100** may be disposed near one or more people to provide a scent experience into the environment.

[0043] FIG. **2** is a perspective view showing a slidable cover **110** of an exemplary digital scent delivery device **100** according to some embodiments. The cover **110** is shown on the first side **101** of the device **100** and on the sides (perpendicular to the first side **101**). As shown, the cover **110** may include a pair of sliders **220** on opposite sides of the cover **110** on an inner surface (i.e., a surface that couples to the device **100**) with a gap **225** between each pair of sliders **220**. The gap **225** between a pair of sliders **220** may fit a guide **230** on the device **100**. Specifically, each side of the housing **115** of the device **100** may include a guide **230**. The device **100** includes two guides **230** disposed on opposite sides of the housing **115**. The guide **230** on each side of the device **100** may be inserted into or withdrawn from a corresponding gap **225** on each side of the cover **110**. In the view shown in FIG. **2**, the sliders **220** and gap **225** are visible on one side of the cover **110** and the guide **230** is visible on an opposite (rather than corresponding) side of the device **100**. Although a slidable cover **110** is shown in discussed with reference to FIG. **2**, any known mechanism may be used to moveably or removably attach the cover **110** according to alternate embodiments. With the cover **110** removed, a replaceable cartridge **210** is visible on the first side **101** of the device **100**. The cartridge **210** may include a number of vessels **215** (generally one or more vessels **215**), each holding a scent medium **310**, as further discussed with reference to FIG. **3**A.

[0044] FIG. **3**A is a perspective view of an exemplary digital scent delivery device **100** according to some embodiments with an outer cover **110** removed. As noted with reference to FIG. **2**, a replaceable cartridge **210** is visible with the outer cover **110** removed. The cartridge **210** may include a number of vessels **215** holding different scent media **310**. The vessels **215** are shown uncovered and most vessels **215** are shown as empty in FIG. **3**A. Each vessel **215** has a first opening **320** and a second opening **330**, as further discussed with reference to FIG. **9**, that are visible for the empty vessels **215**. An exemplary scent medium **310** is shown in one of the vessels **215** in FIG. **3**A. According to different embodiments, two or more of the vessels **215** may include scent media **310**. Non-limiting examples of scent media **310** include foam, wax, paper substrate, liquid, emulsion, gel, polyacrylamide, matrix, electrophoresis, Gellan Gum, cellulose material (e.g., cotton ball, cotton pad), porous polymer, and polysaccharide infused with a particular scent. A scent medium **310** may be molecules of the scent itself. The different scent media **310** may hold different scents for emission or, in alternate embodiments, two or more vessels **215** may include scent media **310** that emit the same scent.

[0045] FIG. **3**B shows the exemplary digital scent delivery device **100** of FIG. **3**A with the cartridge **210** detached from the device **100**. The side **301** of the cartridge **210** that couples to the device **100** is also shown. Exemplary interconnects **305** on the cartridge **210** and device **100** attach the cartridge **210** to the device **100**. In alternate embodiments, different numbers or types of interconnects **305** may be used. Generally, any known approach (e.g., snaping, sliding) to aligning and connecting the cartridge **210** to the device **100** may be used. Each cartridge **210** may be associated with an identifier or code that indicates the available scents and their associated vessels **215**. For example, as indicated in FIG. **3**B, the cartridge **210** may store identification and other information in an electrically erasable programmable read-only memory (EEPROM) **325** and the device **100** may include contact pins **345** to couple to the EEPROM **325** and read information from the EEPROM **325** of the cartridge **210**. Approaches to storing and retrieving information are not limited to the EEPROM **325** and may involve, for example, a radio frequency identifier (RFID) tag or label affixed to the cartridge **210** or a machine readable code aligned with a reader of the device **100**.

[0046] Detaching the cartridge **210** reveals a cartridge interlayer **350** that is the part of the device **100** that couples to the cartridge **210**. The cartridge interlayer **350** includes gaskets **360**, **370**, **380** that correspond with first openings **320**, second openings **330**, and grooves **340** on a side **301** of the cartridge **210** that couples to the device **100**. The gaskets **360**, **370**, **380** (e.g., silicon, rubber, other compliant material) form a seal around the openings **320**, **330** and grooves **340** and may additionally act as a check valve, as further discussed with reference to FIG. **9**.

[0047] The side **301** of the cartridge **210** that couples to the device **100** may be covered with a thin sheet **302** (e.g., film, lamination). The sheet **302** may be coupled to the cartridge **210** via adhesive, ultrasonic or laser bonding, or heat-sealing, for example. In an exemplary embodiment detailed with reference to FIGS. **9** and **11**, the sheet **302** may act as a valve that allows or prevents flow out of the vessels **215** via the second opening **330** of each of the vessels **215**. The sheet **302** may have punctures aligned with the first openings **320** of the vessels **215** such that the sheet **302** does not act as a valve for the first openings **320**. In alternate embodiments, the sheet **302** may additionally or alternately act as a valve for each of the first openings **320**.

[0048] FIG. **3**B includes a view from the perspective of the outlets **120**, looking into the cartridge **210**. This view is shown with the sheet **302** separated from the cartridge **210** to more clearly show the grooves **340** in the body of the cartridge according to exemplary embodiments. When the cartridge **210** includes grooves **340**, a groove **340** associated with a vessel **215** may define part of the channel **920** (FIG. **9**) that facilitates flow of scented gas from the vessel to outside the device **100** (e.g., via an outlet **120**). As discussed with reference to FIG. **11**, for example, the sheet **302** above a particular groove **340** may pull away from the groove **340** based on pressure from exiting scented gas, for example. The sheet **302**, groove **340**, and gasket **380** that seals the sides of the sheet **320** define the channel **920** via which the scented gas flows out of the device **100**.

[0049] FIG. **4** is a perspective view of an exemplary digital scent delivery device **100** according to some embodiments with the housing **115** removed. A controller cover **415** is visible on the first side **101** of the device **100**. A gas supply device **410** may be used to pressurize gas in a reservoir **610** (FIG. **6**) or otherwise cause flow of the gas from the reservoir **610**. As further discussed with reference to FIG. **6**, the gas supply device **410** may be a pressure creator that pressurizes gas in the reservoir **610**. In some embodiments, the gas supply device **410** may be a fan or pump used to actively direct gas out of the reservoir **610**. According to different embodiments, the reservoir **610** may hold gas (e.g., air) or liquified gas (e.g., carbon dioxide). In the exemplary embodiment shown in FIG. **4**, the gas supply device **410** is a piezoelectric-based device **411**.

[0050] FIG. **5** shows the exemplary digital scent delivery device **100** of FIG. **4** with the controller cover **415** removed. The controller **510** generally refers to the various electronics and processing circuitry that may be included to communicate with an external device via wired or wireless communication, identify the cartridge **210** inserted into the device **100** (e.g., based on contact pins **345** contacting an EEPROM **325** of the cartridge **210**), and control scent emission. Communication with the external device may include the controller **510** receiving a message comprised of a character string that indicates a desired scent. As noted with reference to FIG. **3**B, the cartridge **210** that is coupled to the device **100** may include an EEPROM **325** that the controller **510** can read (e.g., via contact pins **345**) to determine the scents and corresponding vessels **215** that are part of the cartridge **210**. The EEPROM **215** may also store the amount of gas that has passed over each medium **310**, the number of scent emission events for each medium **310**, date of filling, and other identifying or use-indicating information. This usage information may be used to replace the cartridge **210**, individual vessels **215**, or their contents. That is, the vessels **215** may be removably placed in the cartridge **210** in addition to or alternately from the cartridge **210** being removably coupled to the device **100**. Additionally or alternately, the vessels **215** may be refillable with scent and/or scented media **310**.

[0051] The controller **510** may map a message string received from an external device to a particular scent (i.e., particular medium **310** in a particular vessel **215**) and control the vessel **215** to emit the scent. The cartridge **210** may indicate (e.g., via information stored in the EEPROM) specific scent mixing formulas. That is, a message received by the controller **510** from an external device may include a string that corresponds with a mixed scent such that the controller **510** controls two or more corresponding vessels **215** to release the mixed scent. The cartridge **210** may additionally limit certain combinations of scent. For example, a manufacturer of a particular scent may specify that a new formulation that includes the scent may not be emitted. In that case, a message received from an external device that specifies the scent in combination with another scent may result in an error message (e.g., red light emitted by light indicator **135**). For example, the controller **510** may issue an error when it receives a string that includes the identifier of the proprietary scent and the identifier of one or more additional scents. In this way, the rights to proprietary scents may be maintained when used in the device **100**.

[0052] FIG. **6** is a cross-sectional view of a first side **101** of an exemplary digital scent delivery device **100** according to some embodiments. The cross-sectional view shows a reservoir **610** within the housing **115** that holds a gas. In the exemplary embodiment shown in FIG. **6**, the gas is air obtained via the intake **620**. In alternate embodiments, the gas may a gas other than air or may be a liquified gas. Structural features (e.g., bracing **615**) may be included in the reservoir **610** to maintain structural integrity of the reservoir **610** and withstand pressure imposed by a pressure creator used as the gas supply device **410** according to some embodiments. In the exemplary case shown in FIG. **6**, the outlet portion of a piezoelectric-based device **411** is shown within the reservoir **610**. As previously noted, a fan or pump may be used, instead, to direct gas (e.g., air, liquified gas) out of the reservoir **610**. The gas in the reservoir **610** may be directed into one or more vessels **215** via corresponding first openings **320**. The scent that is emitted via one or more outlets **120** results from the gas from the reservoir **610** passing over the scent medium **310** within one or more vessels **215** prior to exiting the device **100** via associated one or more outlets **120**, as further discussed with reference to FIG. **9**.

[0053] FIG. **7** is a cross-sectional view of a second side **102** of the exemplary scent delivery device **100** shown in FIG. **6**. The reservoir **610** is shown with the bracing **615** and the gas supply device **410** is shown within the reservoir **610**. As noted, the gas supply device **410** may be a piezoelectric-based device **411** that creates pressure within the reservoir **610** or a fan or pump that urges flow from the reservoir **610**. Gas supply inlets **710** in the cartridge interlayer **350** to supply gas to corresponding vessels **215** of the cartridge **210** are visible in FIG. **7**. While the single reservoir **610** provides gas for all of the gas supply inlets **710** according to the exemplary embodiment shown in FIG. **7**, two or more reservoirs **610** may be used instead, as shown in FIG. **20**. For example, each gas supply inlet **710** may be supplied by a separate reservoir **610**.

[0054] As FIG. **6** indicates, the gas supply inlets **710** on one side of the cartridge interlayer **350** correspond with gaskets **360** on the opposite side of the cartridge interlayer **350**. As indicated by FIG. **3**B, the gaskets **360** are aligned with the first openings **320** in the vessels **215**. The gaskets **360** may act as passive valves at the first openings **320** in the vessels **215**. That is, initially, the passive valves (gaskets **360**) may allow gas from the reservoir **610** to enter the vessels **215** via the first openings **320**. When scent is emitted from a given vessel **215** (i.e., some of the gas allowed into the corresponding vessel **215** is emitted via a corresponding outlet **120**), the gasket **360** corresponding with the given vessel **215** may allow additional gas to enter the vessel **215** via the first opening **320**.

[0055] The gaskets **360** may prevent scented gas in the vessels **215** from flowing back through the first openings **320** and gaskets **360** to the reservoir **610**. In this way, scent contamination (e.g., scent from one vessel **215** entering another vessel **215** via the reservoir **610**) may be prevented. Because all the vessels **215** have gas in them at most times according to the embodiment including passive valves at the first openings **320**, active electromechanical actuation subsystems (e.g., **910**) may be used at the second openings **330** of the vessels **215** to prevent scents from all the vessels **215** being emitted via the outlets **120** at all times.

[0056] According to alternate embodiments, an active electromechanical actuation subsystem may be used at the first openings **320** of the vessels **215** instead. In this case, gas can be prevented from entering the vessels **215** and only a vessel **215** whose scent is to be emitted may have the valve at its first opening **320** controlled to permit entry of pressurized gas. As discussed with reference to FIG. **9**, a passive valve may be used at the second openings **330** in this case. Generally, if a passive valve is used at the first openings **320** or the second openings **330**, an active electromechanical actuation subsystem must be used at the other of the second openings or the first openings **320**. Electromechanical actuation subsystems may be used at both the first openings **320** and the second openings **330**.

[0057] FIG. **8** is a block diagram showing an exemplary interface between a reservoir **610** and a vessel **215** according to some embodiments. According to the view in FIG. **8**, additional vessels **215** may be behind the vessel **215** that is visible. The reservoir **610** shown in FIG. **8** may extend to supply the additional vessels **215** behind the visible vessel **215** or additional reservoirs **610** may be behind the reservoir **610** to supply one or a subset of the vessels **215** behind the visible vessel **215**. The first opening **320** of the vessel **215** is shown at an interface with the reservoir **610**. That is, one or more reservoirs **610** may directly interface with the vessels **215** rather than having the gas supply inlets **710** extend the volume of the reservoir(s) **610** to reach the vessels **215**. As the expanded view illustrates, a sheet **320** (e.g., film, laminate) may be between the reservoir **610** and vessel **215** and have an opening at the first opening **320**, and a gasket **360** may act as a passive valve between the reservoir **610** and vessel **215**. Alternately, there may be an electromechanical actuation subsystem to control flow of gas (e.g., air, liquified gas) from the reservoir **610** into the vessel **215** and prevent backflow of scented gas from the vessel **215** into the reservoir **610**. Alternate interfaces, as compared to the illustrated direct interface or the gas supply inlets **710** of FIG. **7**, between the reservoir(s) **610** and vessels **215** may be used in alternate embodiments.

[0058] FIG. **9** is a cross-sectional view of an exemplary digital scent delivery device **100** according to some embodiments. The side cross-sectional view shown in FIG. **9** illustrates the flow associated with scent emission from one of the vessels **215**. The first opening **320** and second opening **330** of the vessel **215** are indicated. In the exemplary case, the gasket **360** around the first opening **320** acts as a passive valve controlling flow of gas from the reservoir **610** into the vessel **215** and is discussed with reference to FIG. **10**. An electromechanical actuation subsystem **910** is used to control flow of scented gas from the vessel **215** via a channel **920** to the outlet **120** and is discussed with reference to FIG. **11**. The cartridge interlayer **350** is indicated adjacent to the vessel **215**. The gaskets **370**, **380** that surround the second opening **330** and channel **920**, respectively, are not visible in the cross-sectional view.

[0059] FIG. **10** is an expanded view of aspects of the cross-sectional view shown in FIG. **9**. As noted, the exemplary embodiment illustrated in FIGS. **9** and **10** shows the gasket **360** acting as a passive valve at the first opening **320**. Until the vessel **215** is filled, gas that is pressurized in the reservoir **610** or directed via a fan or pump from the reservoir **610**, for example, is allowed to flow from the reservoir **610** into the vessel **215**. The gasket **360** may act as a check valve to prevent the flow of scented gas out of the first opening **320** and back into the reservoir **610**. The gasket **360** acting as a check valve can allow the reservoir **610** to be hyper-pressurized. Any vessel **215** that emits scented gas via the second opening **330** may immediately be re-filled with gas, preventing a delay in any requested scent emission. In this exemplary case, a sheet **302** (e.g., film, laminate) on the cartridge **210**, between the cartridge **210** and cartridge interlayer **350**, may have openings that align with the first openings **320** of the vessels **215** to permit the flow of gas from the reservoir **610** into the vessels **215** via the first openings **320**.

[0060] FIG. **11** is an expanded view of aspects of the cross-sectional view shown in FIG. **9**. As noted, an electromechanical actuation subsystem **910** is used at the second opening **330**. According to an exemplary embodiment, the electromechanical actuation subsystem **910** may include the sheet **302** acting as a valve and a thermoelectric actuator. For example, the sheet **320**, which does not include holes aligned with the second openings **330**, may be a normally closed valve that prevents the out-flow of scented gas from the vessels. The exemplary electromechanical actuation subsystem **910** may include a stopper **1110** that keeps the sheet **302** pressed against the second openings **320**. For example, the stopper **1110** may be in the form of a ceramic ball, as shown in FIG. **11**, positioned at the second opening **330**.

[0061] The actuation subsystem **910** may also include a shape memory alloy **1120** such as a nickel-titanium alloy (Nitinol), further discussed with reference to FIGS. **14**A, **14**B, **15**A and **15**B, that may be positioned to displace the stopper **1110** from the second opening **330**. That is, the shape memory alloy **1120** may contract when a current is applied via the controller **510**. Based on the positioning of the shape memory alloy **1120**, contraction may result in the stopper moving away from the second opening **330** in the direction indicated by the arrow. This may allow gas that entered the vessel **215** via the first opening **320**, which has become scented gas in the vessel **215** due to the scented medium **310**, to flow out of the second opening **330**, through a channel **920**, and out of the device **100** via the outlet **120**. Specifically, as discussed with reference to FIG. **3**B and further detailed in FIG. **11**, the sheet **302** may balloon out away from the second opening **330** based on out-flow of the scented gas from the vessel **215** according to some embodiments, as discussed with reference to FIG. **12**A, for example. The flow of scented gas may be via a channel **920** defined by the sheet **302** and a groove **340** in the cartridge **210**, for example. This and other channel configurations are discussed with reference to FIGS. **12**A, **12**B, and **12**C.

[0062] In alternate embodiments, electrostatic effect (e.g., fluid shape control) or electrodynamic effect (e.g., ion generation) may be used for mechanical actuation of the first openings **320** and/or the second openings **330**. These are further discussed with reference to FIGS. **16-19**. As previously noted, generally one of the openings (first openings **320** or second openings **330**) may be controlled via an electromechanical actuation subsystem. The other openings (second openings **330** or first openings **320**) may have passive valves (e.g., in the form of gaskets **370**, **360**) or electromechanical actuation subsystems to control flow.

[0063] FIGS. **12**A, **12**B, and **12**C illustrate a cross-section along the length from the second openings **330** to the outlets **120**. FIG. **12**A is a view from the perspective of the outlets **120**, similar to the view shown in FIG. **3**B. The grooves **340** in the cartridge **210** are shown and the side **301** of the cartridge that couples to the device **100** is indicated. A sheet **302** covering an open side of the grooves **340** is shown during delivery of scented gas from one of the vessels **215** (second vessel

215 from the left in the example shown). A gasket **380** is on each side of the portion of the sheet **302** that covers each groove **340**, as the illustrations in FIG. **3**B indicate.

[0064] As indicated in FIG. **12**A, when scented gas exits the vessel (second vessel **215** from the left in the example shown), the sheet **302** covering the associated groove **304** js pushed away based on pressure of the exiting scented gas, as indicated by the bump **1210**. The bump **1210** is contained to the area of the sheet **302** over the associated groove **340** by the gasket **380** on either side of the bump **1210**. The bump **1210** and groove **340** form the channel **1220** (e.g., similar to channel **920** in FIGS. **9** and **11**). The sheet **302** covering the associated groove **340** and the associated second opening **330** of the vessel **215** from which the scented gas is output may be pushed away based on an actuation subsystem (e.g., **910**) controlling flow from the second opening **330**. That is, the scented gas is held in the vessel **215** until the second opening **330** is unblocked, at which time the pressure of outflowing scented gas may push the sheet **302** to form the bump **1210** and flow out via the channel **1220**. The second opening **330** may be unblocked based on actuator elements of an actuation subsystem (e.g., **910**, **1405**, **1605**, **1805** (FIGS. **11** and **14**A-**19**)).

[0065] FIG. **12**B is a block diagram that shows a channel **1230** according to an alternate embodiment. In the exemplary embodiment shown in FIG. **12**B, the cartridge **210** does not include grooves **340**. In this case, the scented gas exiting a given vessel **215** pushes the portion of the sheet **302** between its second opening **330** and the outlet **120** away from the surface of the cartridge **210** to form a channel **1230**. As indicated, gaskets **380** may facilitate localizing the disengagement of the sheet **302** from the surface of the cartridge **210**.

[0066] FIG. **12**C is a block diagram that shows a channel **1240** according to another alternate embodiment. In the exemplary embodiment shown in FIG. **12**C, the cartridge **210** includes grooves **340**, but the sheet **302** may not be compliant enough to be pushed away, based on the pressure of exiting scented gas, to form a bump **1210**, as shown for the embodiments of FIGS. **12**A and **12**B.

[0067] In addition, a given path from a second opening **330** to an outlet **120** may include a combination of two or more channels **1220**, **1230**, **1240**. For example, FIG. **3**B, which shows the side **301** of the cartridge **210** that couples to the device **100**, shows that the exemplary cartridge **210** has grooves **340** that do not begin at the second openings **330**. Thus, between a given second opening **330** and corresponding groove **340**, a channel **1230** may be formed, and from the start of the corresponding groove to the corresponding outlet **120**, a channel **1210** (or **1240**) may be formed.

[0068] FIG. **13** is a block diagram of a generalized exemplary vessel **215** of a digital scent delivery device **100** according to some embodiments. The vessel **215** is shown with a scent medium **310** in the vessel **215**. The scent medium **310** may be, for example, a foam, wax, paper substrate, liquid, emulsion, gel, cotton ball, cotton pad. Gas that enters the vessel **215** becomes scented gas. The first opening **320** for entry of gas (e.g., air, liquified gas) into the vessel **215** is indicated with a first valve **1310** controlling the flow into the first opening **320**. The first valve **1310** may be a passive valve formed by a sheet **302** and/or gasket **360**, for example. In alternate embodiments, the first valve **1310** may be part of an actuation subsystem according to any of the embodiments discussed herein. In this case, the first valve **1310** may be closed by default and opened based on control of an actuator. For example, as shown in FIG. **9**, an exemplary actuation subsystem **910** involves a sheet **302** acting as a valve and an electromechanical actuator employing thermoelectric, electrostatic effect, or electrodynamic effect to control opening of the first opening **320**. The first valve **1310** prevents scented gas in the vessel **215** from exiting via the first opening **320**. This ensures that the reservoir **610** is not contaminated with scent and does not subsequently cross-contaminate the scent in another vessel **215**, for example.

[0069] The second opening **330** for exit of scented gas from the vessel **215** is indicated with a second valve **1320** controlling the flow out of the second opening **330**. As discussed with reference to the first valve **1310**, the second valve **1320** may be a passive valve or may be part of an actuation subsystem. One or both of the first valve **1310** or the second valve **1320** may be part of an actuation subsystem. When both the first valve **1310** and the second valve **1320** are part of an actuation subsystem, they may be part of the same or a different type of actuation subsystem (e.g., thermoelectric, electrostatic, electrodynamic).

[0070] FIGS. **14**A, **14**B, **15**A and **15**B are used to illustrate closed and open states of the first opening **320**, second opening **330**, or outlet path **1400** based on an exemplary actuation subsystem **1405** employing thermoelectric effect. Although the system may employ thermoelectric effect, it should be appreciated that in some embodiments, any electromechanical elements may be used including piezoelectric devices or other type of element. The outlet path **1400** refers to any part of the path between a second opening **330** and a corresponding outlet **120**. The actuation subsystem **1405** may include a valve **1410**. The valve **1410** may be the sheet **302**, gasket **360** (if covering the first opening **320**), gasket **370** (if covering the second opening **330**), gasket **380** (if covering the outlet path **1400**) or other component that blocks flow of gas into the vessel **215**, flow of scented gas out of the vessel **215**, or flow of scented gas out of the device **100** as shown in FIG. **14**A. For example, in the case of the valve **1410** over the outlet path **1400**, the valve **1410** may be the sheet **302** above the surface of the cartridge **210** (as in FIG. **12**B) or above a groove **340** in the cartridge **210**. In some embodiments, the valve **1410** may be a combination of the sheet **302** and gasket **360**, **370**, **380**, as shown in FIG. **14**B. As noted with reference to FIG. **3**B, the sheet **302** may be a film. Thus, the exemplary embodiment shown in FIG. **14**B may involve a film and gasket **360**, **370**, or **380** covering the first opening **320**, second opening **330**, or outlet path **1400**.

[0071] The actuation subsystem **1405** may also include actuation elements. For example, a thermoelectric actuator may include a shape memory alloy **1420** (e.g., nitinol) and a stopper **1430** (e.g., ceramic ball stopper **1110**) positioned to keep the valve **1410** in the closed position, illustrated in FIGS. **14**A and **14**B, as a default. Based on application of current to the shape memory alloy **1420** (e.g., via controller **510** operation), the shape memory alloy **1420** may deform, as shown in FIGS. **15**A and **15**B. As a result, the stopper **1430** may move, as shown, and open the first opening **320**, second opening **330**, or outlet path **1410** (e.g., channel **1220**, **1230**, **1240**). While the exemplary actuation subsystem **1405** operates based on thermoelectric effect, alternate embodiments may, instead, operate based on electrostatic, electrodynamic, or reverse piezoelectric effect to cause mechanical motion based on an electric control signal. Piezoelectric and electrostatic electromechanical actuation subsystems **1605**, **1805** are discussed with reference to FIGS. **16-19**. According to exemplary embodiments, as discussed with reference to FIGS. **16-19**, the actuator may act directly on the valve **1410** rather than indirectly, by acting on a stopper **1430**.

[0072] FIGS. **16** and **17** are used to illustrate closed and open states of the first opening **320**, second opening **330**, or outlet path **1400** based on an exemplary actuation subsystem **1605** employing piezoelectric effect. As noted, another variation shown in FIGS. **16** and **17**, as compared with FIGS. **14**A, **14**B, **15**A, and **15**B, is the lack of a separate stopper and actuator. That is, FIGS. **16** and **17** show a piezoelectric element **1610** that blocks a valve **1410** from opening, according to the illustration in FIG. **16**, and that deforms based on application of current to allow the valve **1410** to open, as illustrated in FIG. **17**. In alternate embodiments, a separate stopper may be used with the piezoelectric element **1610** for operation similar to the embodiment shown in FIGS. **9**, **11**, **14**A, **14**B, **15**A, and **15**B.

[0073] FIGS. **18** and **19** are used to illustrate closed and open states of the first opening **320**, second opening **330**, or outlet path **1400** based on an exemplary actuation subsystem **1805** employing electrostatic effect. As shown, the actuator **1810**, which also acts as a stopper to keep the valve **1410**, as illustrated in FIG. **18**, includes two surfaces **1815***a*, **1815***b* that hold a (dielectric) liquid **1820** therebetween. When current is applied, the surfaces **1815***a*, **1815***b* flatten, thereby allowing the valve **1410** to open, as shown in FIG. **19**. As noted, in additional alternate embodiments, electrodynamic effect may be used to control a valve **1410**.

[0074] FIG. **20** is a block diagram of a digital scent delivery device **100** according to some embodiments. The scent delivery device **100** in FIG. **20** is a generalized version of the scent delivery device **100** shown according to exemplary embodiments in prior figures. A controller **10** may communicate with an external device **15** to select one or more scents for delivery. Each scent may be stored in a scent medium **310** in a vessel **30**. Each vessel **30** may include a first opening valve **25** (e.g., first opening **320** and passive valve or actuation subsystem) that controls input of a gas (e.g., air, liquified gas) into the vessel **30** from a corresponding reservoir **20**.

[0075] In the exemplary digital scent delivery device **100** shown in FIG. **20**, each vessel **30** is supplied by a corresponding reservoir **20**. Each reservoir **20** has an associated gas moving device **22** (e.g., fan, air pump) as the gas supply device **410** according to the exemplary embodiment. The controller **10** may control the gas moving device **22**. The controller **10** may also communicate with an EEPROM **40** or other memory that stores information about the scent stored by each vessel **30** and other information (e.g., usage information).

[0076] Each vessel **30** includes a second opening valve **35** (e.g., second opening **330** and passive valve or actuation subsystem) that controls output of scented gas from the vessel **30**. The digital scent delivery device **100** may include a mixing area **50** according to some embodiments. When a combination of scents is delivered, the scents may be combined in an optional mixing area **50** prior to being released from the digital scent delivery device **100** via one or more outlets **120**.

[0077] FIG. **21** is a block diagram of a digital scent delivery device **100** according to some embodiments. The scent delivery device **100** in FIG. **21** is a generalized version of the scent delivery device **100** shown according to exemplary embodiments in prior figures and shows some variations from the features shown in FIG. **20**. For example, the exemplary embodiment shown in FIG. **21** does not include a mixing area **50**. In this type of embodiment, a deflector may be added at one or more outlets **120** to deflect output of the scent medium **310** in one or more of the vessels **30**. For example, a deflector (e.g., plastic) may be added at the outlets **120** on the two ends to direct the scent media **310** from those outsets **120** closer to the center. Deflectors may be used at one or more outlets **120** to control directionality of scent for any reason. Additionally or alternately, a fan or other amplification device may be added at one or more outlets **120** to increase the flow velocity from one or more vessels **30** or the mixing area **50**. In some embodiments, the controller **10** may modulate intensity of one or more scents by controlling the flow actuation and/or amplification. For example, the EEPROM **325** or other storage associated with the cartridge **210** may specify scent intensity or scent mixing by weight for different combinations of scents. It should be understood that the features shown and discussed with reference to one or more figures may be combined according to contemplated alternate embodiments.

[0078] The exemplary embodiment of FIG. **21** shows one reservoir **20** for supply of gas to each of the vessels **30**. The reservoir **20** may have a pressure creator **60** (e.g., piezoelectric-based device) as the gas supply device **410**. A pressure sensor **65** may monitor pressure in the reservoir **60** and communicate with the controller **10** to facilitate control of the pressure via control of the pressure creator **60** by the controller **10** based on the sensed pressure. In some embodiments, data from the pressure sensor **65** may be used to guide functionality. For example, if the pressure sensor **65** indicates that pressure in the reservoir **60** is below a threshold value, scent delivery may be paused/delayed or another component (e.g., piezoelectric-based device **411**, external device) may be signaled to increase pressure in the reservoir **60**. The reservoir **20** is shown with dashed lines to indicate that a reservoir **20** as such may not be used at all according to exemplary embodiments. Instead, for example, air may be pulled into the device **100** and supplied (e.g., with or without pressurization) to the vessels **30**.

[0079] It should be appreciated that there are other applications of this technology and the invention is not limited to the examples provided herein. The device may be incorporated in part or whole into other devices or connected to other devices. For example, some embodiments may be used in general entertainment, which could be movies or other experiences. Additionally, some embodiments may be applied to areas such as travel, business, education/training, telepresence, and meditation.

[0080] Having thus described several aspects of at least one embodiment of this invention, it is to be appreciated various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of this disclosure, and are intended to be within the spirit and scope of the invention. Accordingly, the foregoing description and drawings are by way of example only.

Claims

1. A system comprising: a plurality of vessels, each of the plurality of vessels including a scented medium between a first opening and a second opening; a first valve disposed at the first opening of each of the plurality of vessels; a second valve disposed at the second opening of each of the plurality of vessels, wherein for each of the plurality of vessels, at least one of the first valve or the second valve is part of a first electromechanical actuation subsystem and another of the first valve or the second valve is a passive valve or is part of a second electromechanical actuation subsystem; one or more reservoirs holding a gas; one or more interconnections arranged to supply the gas from the one or more reservoirs to the first opening of the plurality of vessels; a controller configured to control release of scented gas from at least one of the plurality of vessels via the second opening of the at least one of the plurality of vessels to a corresponding outlet.
2. The system of claim 1, wherein the scented gas results from the gas from one of the one or more reservoirs interacting with the scented medium in the at least one of the plurality of vessels.
3. The system of claim 1, wherein the scented medium is foam, wax, a paper substrate, gel, cotton ball, cotton pad, or porous polymer.
4. The system of claim 1, wherein the first electromechanical actuation subsystem and the second electromechanical actuation subsystem are a same type of electromechanical actuation subsystem.
5. The system of claim 1, wherein at least one of the first electromechanical actuation subsystem or the second electromechanical subsystem includes a thermoelectric actuator.
6. The system according to claim 5, wherein the thermoelectric actuator includes a shape memory alloy configured to move based on an application of a current and a stopper configured to cover at least one of the first opening or the second opening until movement of the shape memory alloy moves the stopper.
7. The system of claim 5, wherein the thermoelectric actuator includes Nitinol.
8. The system of claim 1, wherein at least one of the first electromechanical actuation subsystem or the second electromechanical actuation subsystem operates based on an electrostatic or electrodynamic effect.
9. The system of claim 1, wherein at least one of the first electromechanical actuation subsystem or the second electromechanical actuation subsystem includes a piezoelectric actuator.
10. The system of claim 1, wherein the gas is pressurized air.
11. The system of claim 1, wherein the gas is liquified gas.
12. The system of claim 11, wherein the liquified gas is carbon dioxide.
13. The system of claim 1, further comprising a fan or air pump coupled to the one or more reservoirs to urge flow of the gas from the one or more reservoirs.
14. The system of claim 1, wherein the controller is configured to communicate with an external device and control release of the scented gas based on communication from the external device.
15. The system of claim 14, wherein the controller is configured to communicate with the external device wirelessly or via a port.
16. The system of claim 1, further comprising a cartridge configured to house the plurality of vessels.
17. The system of claim 16, further comprising a cartridge interface, wherein the cartridge is configured to detach from the system and the cartridge interface is configured to couple to the cartridge when the cartridge is attached to the system.

18. The system of claim 17, wherein the cartridge interface includes first openings configured to align, respectively, to the first openings of the plurality of vessels housed in the cartridge, and first gaskets configured to seal an interface between the first openings of the cartridge interface and the first openings of the plurality of vessels, respectively.
19. The system of claim 17, wherein the cartridge includes grooves defining each channel from the second opening of each of the plurality of vessels to each corresponding outlet and the cartridge interface includes channel gaskets configured to seal each of the channels.
20. The system of claim 17, wherein the cartridge includes a sheet covering a side of the cartridge that couples to the cartridge interface.
21. The system of claim 20, wherein each channel is between the sheet and one of the grooves.
22. The system of claim 19, wherein the sheet is a film or a laminate.
23. The system of claim 16, further comprising a removable cover configured to cover the cartridge.
24. The system of claim 16, wherein the controller obtains information from the cartridge.
25. The system of claim 24, wherein the information identifies a scent stored in each of the plurality of vessels.
26. The system of claim 24, wherein the information is stored in an electrically erasable programmable read-only memory (EEPROM) of the cartridge.
27. The system of claim 24, wherein the information indicates allowed and disallowed combinations of scents.
28. The system of claim 24, wherein the information indicates usage of one or more of the plurality of vessels.
29. The system of claim 24, wherein the information is used to replace the cartridge.
30. The system of claim 24, wherein the information is used to refill or replace one or more of the plurality of vessels.